Activity 5: Technology Architecture Design - Sample Solution

IT Asset Management Transformation Using Snipe-IT

For: PUP IT Asset Management System (PUP-ITAMS)

Executive Summary

This sample solution demonstrates the complete technology architecture for the PUP IT Asset Management System transformation. The architecture is built on AWS cloud services with a polyglot persistence strategy, event-driven AI integration, and infrastructure as code. All decisions are traced back to specific pain points identified in Activity 1 and TO-BE processes from Activity 2.

Assumptions for this solution:

- Company: PUP-ITAMS (serving 5,000+ faculty/staff, 15,000+ students)
- Cloud Provider: AWS (ap-southeast-1 / Singapore region)
- Primary Software: Snipe-IT (customized)
- Complexity Tier: Tier 2 (Intermediate)

Deliverable 1: Deployment Architecture Diagram

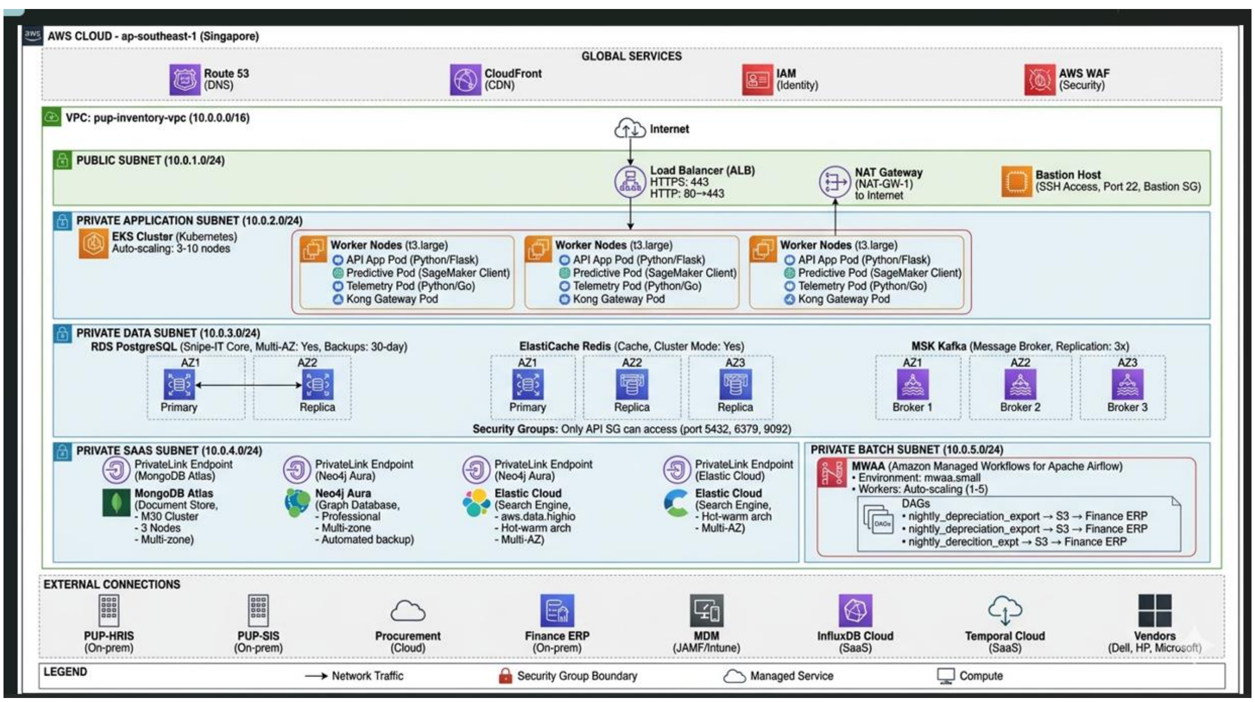
Container to Infrastructure Mapping

Container	Deployment Target	Instance Type	Scaling	Availability Zone	Data Residency
Web Application	AWS S3 + CloudFront	Static hosting	Global CDN	Multi-region	Singapore
Mobile App	App Store / Play Store	N/A	N/A	Global	N/A

Container	Deployment Target	Instance Type	Scaling	Availability Zone	Data Residency
API Gateway	Kong on EKS	Managed	Auto	Multi-AZ	Singapore
API Application	AWS EKS	t3.large	Horizontal (3-10 pods)	Multi-AZ	Singapore
PostgreSQL (Snipe-IT)	AWS RDS	db.t3.large	Vertical + Read replicas	Multi-AZ	Singapore
MongoDB	MongoDB Atlas	M30	Horizontal sharding	Multi-zone	Singapore
InfluxDB	InfluxDB Cloud	Starter tier	Auto-scaling	Regional	Singapore
Neo4j	Neo4j Aura	Professional	Vertical	Regional	Singapore
Redis	AWS ElastiCache	cache.t3.micro	Cluster mode	Multi-AZ	Singapore
Kafka	AWS MSK	kafka.t3.small	Auto-scaling	Multi-AZ	Singapore
Elasticsearch	Elastic Cloud	aws.data.highio	Hot-warm	Multi-AZ	Singapore
Debezium	Kafka Connect	Same as Kafka	Managed connector	Multi-AZ	Singapore

Container	Deployment Target	Instance Type	Scaling	Availability Zone	Data Residency
Predictive Service	AWS EKS	t3.medium	Horizontal (2-5 pods)	Multi-AZ	Singapore
Telemetry Service	AWS EKS	t3.medium	Horizontal (2-5 pods)	Multi-AZ	Singapore
Workflow Engine	Temporal Cloud	Standard	Managed	Multi-region	Singapore
Batch Processor	AWS MWAA	mwaa.small	Auto-scaling	Single-AZ	Singapore

Deployment Diagram Description

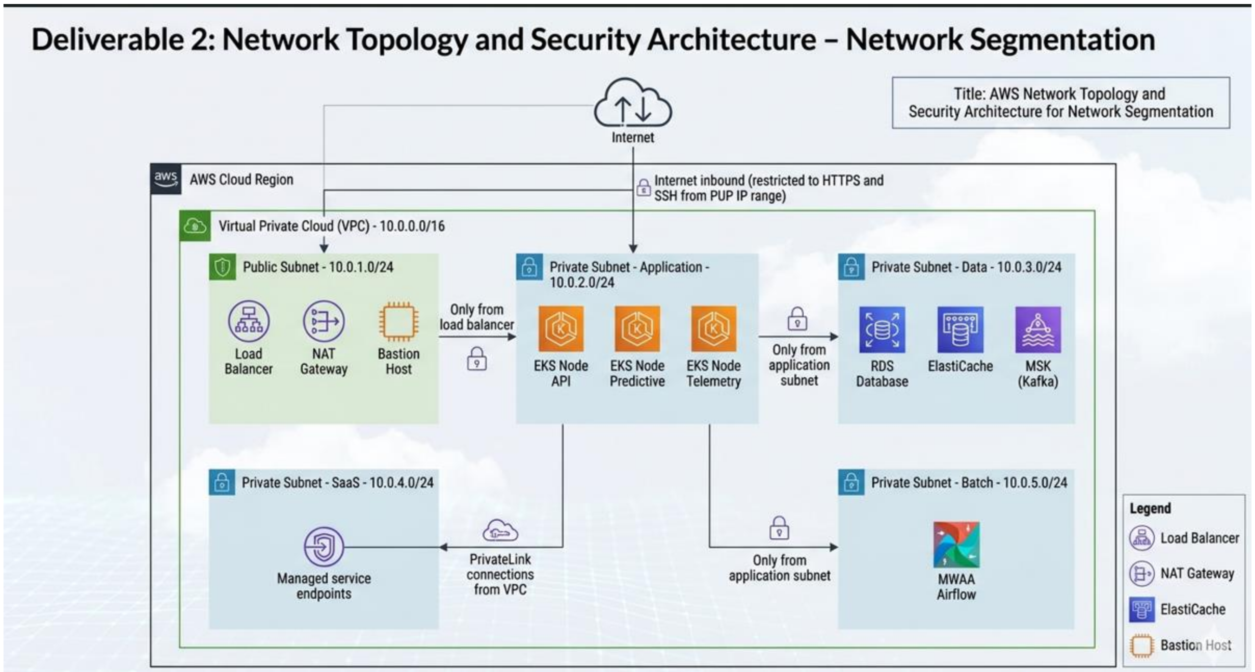


The architecture is deployed entirely within the AWS ap-southeast-1 (Singapore) region to serve the Philippine university. The VPC is segmented into five subnets:

- **Public Subnet (10.0.1.0/24):** Contains the internet-facing load balancer, NAT gateways for outbound internet access, and a bastion host for administrative SSH access.
- **Private Application Subnet (10.0.2.0/24):** Hosts the EKS cluster nodes running the API application, Predictive Service, and Telemetry Service. This subnet has no direct internet access.
- **Private Data Subnet (10.0.3.0/24):** Contains the RDS PostgreSQL instance, ElastiCache Redis cluster, and MSK Kafka brokers. Security groups restrict access to only the application subnet.
- **Private SaaS Subnet (10.0.4.0/24):** Provides PrivateLink connections to MongoDB Atlas, Neo4j Aura, and Elastic Cloud, allowing traffic to remain within AWS network.
- **Private Batch Subnet (10.0.5.0/24):** Hosts the MWAA Airflow environment for batch processing jobs.

The EKS cluster is configured with auto-scaling node groups that scale from 3 to 10 nodes based on CPU and memory metrics. The RDS PostgreSQL instance is Multi-AZ with a standby replica in a different availability zone. All managed SaaS services (MongoDB Atlas, InfluxDB Cloud, Neo4j Aura, Elastic Cloud, Temporal Cloud) are configured with their respective high-availability options.

Deliverable 2: Network Topology and Security Architecture



Network Segmentation

Network Layer	CIDR	Purpose	Access
Public Subnet	10.0.1.0/24	Load balancers, NAT gateways, bastion hosts	Internet inbound (restricted to HTTPS and SSH from PUP IP range)
Private Subnet - Application	10.0.2.0/24	EKS nodes (API, Predictive, Telemetry)	Only from load balancer
Private Subnet - Data	10.0.3.0/24	RDS, ElastiCache, MSK	Only from application subnet

Network Layer	CIDR	Purpose	Access
Private Subnet - SaaS	10.0.4.0/24	Managed service endpoints	PrivateLink connections from VPC
Private Subnet - Batch	10.0.5.0/24	MWAA Airflow	Only from application subnet

Security Group Rules

Security Group	Inbound Rules	Outbound Rules	Description
Load Balancer SG	HTTPS (443) from 0.0.0.0/0, HTTP (80) redirect to HTTPS	HTTPS to API SG	Internet-facing traffic
API SG	HTTPS (443) from Load Balancer SG, SSH (22) from Bastion SG	HTTPS to internet (external APIs), PostgreSQL (5432) to Data SG, Redis (6379) to Data SG	API application instances
Database SG	PostgreSQL (5432) from API SG, Redis (6379) from API SG, Kafka (9092) from API SG	None (locked down)	Data layer components
Bastion SG	SSH (22) from PUP corporate IP range (203.177.x.x)	SSH to API SG, Database SG	Administrative access

Security Group	Inbound Rules	Outbound Rules	Description
SaaS Endpoint SG	PrivateLink connections from VPC	None	MongoDB Atlas, Neo4j Aura, Elastic Cloud

Data Encryption Strategy

Data Category	At Rest Encryption	In Transit Encryption	Key Management
RDS PostgreSQL	AWS RDS encryption (AES-256)	TLS 1.2+	AWS KMS
MongoDB Atlas	Atlas encryption (AES-256)	TLS 1.2+	Cloud provider KMS
InfluxDB Cloud	Platform encryption	TLS 1.2+	Cloud provider
Neo4j Aura	Platform encryption	TLS 1.2+	Cloud provider
S3 Static Assets	S3-SSE (AES-256)	TLS 1.2+	AWS KMS
EBS Volumes	AWS EBS encryption	N/A	AWS KMS
API Traffic	N/A	TLS 1.2+	AWS Certificate Manager

Data Category	At Rest Encryption	In Transit Encryption	Key Management
Backups	Encrypted snapshots	TLS 1.2+	AWS KMS

Deliverable 3: Technology Stack Selection Matrix

Component	Technology	Version	Justification	Alternatives Considered	Selection Rationale
Cloud Provider	AWS	-	Broadest service offering, mature AI/ML services, strong regional presence in Singapore	Azure, GCP	AWS has best SageMaker for AI, RDS for PostgreSQL, and MSK for Kafka
Web Frontend	React.js	18.x	Strong ecosystem, team expertise, component reusability	Angular, Vue	Largest talent pool, excellent performance
Mobile App	React Native	0.72	Code reuse with web, large community, good for scanning apps	Flutter, Native	Shared codebase reduces development effort

API Framework	Python/FastAPI	0.100	Rapid development, excellent ML/AI libraries, async support	Flask, Django	FastAPI provides automatic OpenAPI docs and async support
API Gateway	Kong (on EKS)	3.x	Open source, plugin ecosystem, no vendor lock-in	AWS API Gateway	Avoids vendor lock-in, runs on same EKS cluster
Container Orchestration	EKS (Kubernetes)	1.28	Portability, ecosystem, auto-scaling, service discovery	ECS, Fargate	Kubernetes skills transferable; EKS managed control plane
Core Database	RDS PostgreSQL	14+	Managed ACID compliance, automated backups, Multi-AZ	Aurora PostgreSQL	RDS is cost-effective; Aurora premium not justified
Document Store	MongoDB Atlas	6+	Managed document DB, flexible schema, global clusters	DynamoDB, Couchbase	Atlas avoids vendor lock-in; free tier available
Time-Series DB	InfluxDB Cloud	2.x	Optimized for telemetry, high	TimescaleDB, Prometheus	Better compression and query

			write throughput		language for time-series
Graph Database	Neo4j Aura	5.x	Best graph query performance, managed service	Amazon Neptune	Free tier for development; avoids vendor lock-in
Cache	ElastiCache Redis	7.x	Managed Redis, cluster mode, high performance	Memcached	Redis supports data structures needed for rate limiting
Message Broker	MSK (Kafka)	3.x	Managed Kafka, high throughput, durability	RabbitMQ, SQS	Kafka needed for CDC and event sourcing
CDC	Debezium on MSK Connect	2.x	Native PostgreSQL support, Kafka integration	Custom CDC	Debezium is standard for CDC; managed connector
Search Engine	Elastic Cloud	8.x	Managed Elasticsearch, full-text search	OpenSearch	Better developer experience and features
Workflow Engine	Temporal Cloud	Latest	Managed workflow engine,	Airflow, Camunda	Designed for long-running

			durable execution		stateful workflows
Batch Processing	MWAA (Airflow)	2.x	Managed Airflow, DAG scheduling, Python-native	Glue, Step Functions	Airflow is standard for batch ETL
AI Model Serving	SageMaker	-	Managed endpoints, auto-scaling, monitoring	Custom container, Lambda	Handles model versioning, A/B testing, monitoring
Infrastructure as Code	Terraform	1.5	Cloud-agnostic, mature state management	Pulumi, CloudFormation	Largest provider ecosystem, team familiarity
CI/CD	GitLab CI	-	Integrated with source control, flexible pipelines	Jenkins, GitHub Actions	Self-hosted option, integrated with our GitLab instance
Monitoring	Prometheus + Grafana	-	Industry standard, flexible metrics, open source	Datadog, CloudWatch	No vendor lock-in, large ecosystem

Logging	ELK Stack	8.x	Centralized logging, full-text search, visualization	Splunk, Loki	Elastic Cloud provides managed service
Tracing	Jaeger	1.50	OpenTelemetry compatible, distributed tracing	Zipkin	Better features, active development

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Deliverable 4: Managed Services Selection and Justification

Managed Services Decision Matrix

Service Type	Self-Hosted Alternative	Managed Service	Selection	Rationale
PostgreSQL	Self-managed on EC2	RDS	✓ Managed	Automated backups, Multi-AZ failover, patch management, monitoring
MongoDB	Self-managed on EC2	MongoDB Atlas	✓ Managed	Automatic sharding, backups, global clusters, reduces DBA overhead
Redis	Self-managed on EC2	ElastiCache	✓ Managed	Automatic failover, backup/restore, monitoring, patching
Kafka	Self-managed on EC2	MSK	✓ Managed	Automated provisioning, Apache Kafka compatible, reduced ops

Service Type	Self-Hosted Alternative	Managed Service	Selection	Rationale
Airflow	Self-managed on EC2	MWAA	✓ Managed	Managed workers, automatic scaling, security patching
Kubernetes	Self-managed K8s	EKS	✓ Managed	Managed control plane, updates, security patches
InfluxDB	Self-managed on EC2	InfluxDB Cloud	✓ Managed	Serverless, automatic scaling, built-in downsampling
Neo4j	Self-managed on EC2	Neo4j Aura	✓ Managed	Automated backups, scaling, security updates
Elasticsearch	Self-managed on EC2	Elastic Cloud	✓ Managed	Automated upgrades, security, snapshot management
Temporal	Self-managed on EC2	Temporal Cloud	✓ Managed	Durable execution, scaling, monitoring

AI/ML Managed Services

Service	Purpose	Managed Offering	Key Features
Model Training	Demand forecasting model	SageMaker Training	Managed training instances, automatic model tuning, distributed training

Service	Purpose	Managed Offering	Key Features
Model Hosting	Real-time predictions	SageMaker Endpoints	Auto-scaling, A/B testing, model monitoring, explainability
Data Labeling	Damage detection training	SageMaker Ground Truth	Active learning, workforce management, quality control
Feature Store	ML features for forecasting	SageMaker Feature Store	Online/offline storage, feature versioning, consistency

Cost vs. Operational Overhead Analysis

Component	Self-Hosted Monthly Cost	Managed Monthly Cost	Operational Hours Saved (month)	Recommendation
PostgreSQL	~$150 (EC2 + storage)	~$250 (RDS)	20 hours	Managed
MongoDB	~$200 (EC2 cluster)	~$300 (Atlas)	15 hours	Managed
Kafka	~$300 (3 brokers)	~$400 (MSK)	25 hours	Managed
Kubernetes	~$200 (control plane)	~$75 (EKS)	30 hours	Managed

Component	Self-Hosted Monthly Cost	Managed Monthly Cost	Operational Hours Saved (month)	Recommendation
Total	**~$850**	**~$1,025**	**90 hours**	**Managed**

Conclusion: The additional 175 per month is justified by saving 90 engineer hours monthly, valued at approximately 9,000 (assuming $100/hour engineer cost).

Deliverable 5: Infrastructure as Code Strategy

IaC Tool Selection

Tool	Purpose	Selection Rationale
Terraform	Primary IaC for all AWS resources	Cloud-agnostic, mature state management, large provider ecosystem, team familiarity
Terragrunt	Terraform wrapper for multiple environments	DRY configuration, remote state management, environment segregation
GitHub Actions	CI/CD for infrastructure changes	Integrated with source control, pull request workflows, cost-effective

Terraform Configuration Structure

```
terraform/
 ├── environments/
 |   ├── dev/
 |   |   ├── main.tf
```

```
│   │   ├── variables.tf
│   │   └── terraform.tfvars
│   ├── staging/
│   │   ├── main.tf
│   │   ├── variables.tf
│   │   └── terraform.tfvars
│   └── prod/
│       ├── main.tf
│       ├── variables.tf
│       └── terraform.tfvars
├── modules/
│   ├── networking/
│   │   ├── main.tf
│   │   ├── variables.tf
│   │   └── outputs.tf
│   ├── compute/
│   │   ├── main.tf
│   │   ├── variables.tf
│   │   └── outputs.tf
│   ├── database/
│   │   ├── main.tf
│   │   ├── variables.tf
│   │   └── outputs.tf
│   └── security/
│       ├── main.tf
```

```
|       ├── variables.tf
|      └── outputs.tf
└── global/
    ├── iam/
    ├── s3-backend/
   └── route53/
```

Terraform Configuration Example

hcl

```
# modules/database/rds/main.tf
resource "aws_db_instance" "inventory_db" {
  identifier     = "inventory-db-${var.environment}"
  engine         = "postgres"
  engine_version = "14.9"
  instance_class = var.instance_class

  allocated_storage     = var.allocated_storage
  storage_encrypted     = true
  storage_type          = "gp3"

  db_name  = var.db_name
  username = var.db_username
  password = random_password.db_password.result

  vpc_security_group_ids = [aws_security_group.rds.id]
  db_subnet_group_name   = aws_db_subnet_group.main.name
```

```
  backup_retention_period = var.backup_retention_days
  backup_window        = "03:00-04:00"
  maintenance_window     = "sun:04:00-sun:05:00"

  multi_az             = var.environment == "prod"
  skip_final_snapshot    = var.environment != "prod"

  tags = {
    Name       = "inventory-db-${var.environment}"
    Environment = var.environment
    ManagedBy   = "terraform"
  }
}

# environments/prod/terraform.tfvars
environment        = "prod"
region             = "ap-southeast-1"
vpc_cidr           = "10.0.0.0/16"
db_instance_class   = "db.t3.large"
db_allocated_storage = 100
backup_retention_days = 30
```

Infrastructure CI/CD Pipeline

Stage	Action	Tool	Gate
Plan	terraform plan on PR	GitHub Actions	Review comment
Review	Manual approval for production	GitHub PR	2 approvals required
Apply (Dev)	Auto-apply on merge to dev	GitHub Actions	Tests pass
Apply (Staging)	Auto-apply on tag	GitHub Actions	Dev deploy successful
Apply (Prod)	Manual trigger with approval	GitHub Actions	Staging verified

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Deliverable 6: Scalability and High Availability Design

Component Scalability Matrix

Component	Scaling Type	Scaling Trigger	Min Instances	Max Instances	Scale-up Time
API Application	Horizontal (HPA)	CPU > 70%, Memory > 80%	3	20	2-3 minutes
Predictive Service	Horizontal (HPA)	CPU > 60%, Queue length > 100	2	10	2-3 minutes

Component	Scaling Type	Scaling Trigger	Min Instances	Max Instances	Scale-up Time
Telemetry Service	Horizontal (HPA)	CPU > 70%, Messages/sec > 1000	2	15	2-3 minutes
PostgreSQL	Vertical	Read replica count, storage	1 primary + 1 replica	1 + 5 replicas	10-15 minutes
MongoDB	Horizontal (sharding)	Storage > 80%, Write throughput	3 nodes	Unlimited	5-10 minutes
Kafka	Horizontal (partitions)	Partition lag > 10000	3 brokers	12 brokers	5 minutes
Redis	Cluster (sharding)	Memory > 75%	1 primary + 2 replicas	15 shards	5 minutes

High Availability Configuration

Component	HA Strategy	RTO (Recovery Time)	RPO (Recovery Point)	SLA
API Application	Multi-AZ deployment, load balancer	< 1 minute	0 (stateless)	99.95%
PostgreSQL	Multi-AZ with automatic failover	< 2 minutes	< 5 seconds (sync replica)	99.95%

Component	HA Strategy	RTO (Recovery Time)	RPO (Recovery Point)	SLA
MongoDB	Multi-zone cluster (3 nodes)	< 1 minute	0 (majority write)	99.99%
Kafka	Multi-AZ, replication factor 3	< 2 minutes	0 (acks=all)	99.9%
Redis	Cluster with replicas in each AZ	< 1 minute	< 1 second	99.9%
InfluxDB	Cloud-managed multi-zone	< 2 minutes	< 1 minute	99.95%
Neo4j	Cloud-managed multi-zone	< 2 minutes	< 5 minutes	99.95%

Disaster Recovery Plan

Tier	Recovery Objective	Strategy	Annual Testing
Critical (Order processing, inventory)	RTO < 1 hour, RPO < 5 minutes	Cross-region replication (ap-southeast-2), active-passive	Quarterly
Important (User data, assignments)	RTO < 4 hours, RPO < 1 hour	Daily backups to separate region (ap-southeast-2), point-in-time recovery	Semi-annual

Tier	Recovery Objective	Strategy	Annual Testing
Normal (Logs, telemetry)	RTO < 24 hours, RPO < 24 hours	Weekly backups to S3 with cross-region replication	Annual

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Deliverable 7: Cost Estimation and Budget Analysis

Monthly Cost Breakdown

Service	Instance Type	Quantity	Unit Price (USD)	Monthly Total (USD)	Annual Total (USD)
Compute					
EKS Cluster	Control plane	1	$72	$72	$864
EKS Worker Nodes (API)	t3.large (spot)	3	$35	$105	$1,260
EKS Worker Nodes (Predictive)	t3.medium (spot)	2	$17.50	$35	$420
EKS Worker Nodes (Telemetry)	t3.medium (spot)	2	$17.50	$35	$420
Databases					

Service	Instance Type	Quantity	Unit Price (USD)	Monthly Total (USD)	Annual Total (USD)
RDS PostgreSQL	db.t3.large (Multi-AZ)	1	$150	$150	$1,800
RDS Storage	gp3, 100GB	1	$10	$10	$120
MongoDB Atlas	M30 (3 nodes)	1	$300	$300	$3,600
InfluxDB Cloud	Starter tier	1	$49	$49	$588
Neo4j Aura	Professional	1	$199	$199	$2,388
ElastiCache Redis	cache.t3.micro (cluster)	3	$12	$36	$432
Integration					
MSK Kafka	kafka.t3.small (3 brokers)	1	$150	$150	$1,800
MSK Storage	100GB per broker	3	$10	$30	$360
MWAA Airflow	mwaa.small	1	$150	$150	$1,800
AI/ML					

Service	Instance Type	Quantity	Unit Price (USD)	Monthly Total (USD)	Annual Total (USD)
SageMaker Endpoint	ml.t3.medium	1	$90	$90	$1,080
SageMaker Training	ml.t3.large (occasional)	-	$20	$20	$240
Networking					
NAT Gateway	-	2	$32	$64	$768
Data Transfer	1 TB	-	$90	$90	$1,080
CloudFront	1 TB	-	$85	$85	$1,020
Load Balancer	ALB	1	$22	$22	$264
Storage & Backup					
S3 (static assets)	10 GB	1	$0.23	$0.23	$2.76
RDS Backup Storage	100 GB	1	$10	$10	$120
Management & Security					

Service	Instance Type	Quantity	Unit Price (USD)	Monthly Total (USD)	Annual Total (USD)
AWS WAF	-	1	$25	$25	$300
TOTAL				$4,927.23	$59,126.76

Cost Optimization Strategies

Strategy	Description	Estimated Savings	Priority
Spot Instances	Use spot instances for non-critical workloads (batch, dev, telemetry)	60-70% on compute	High
Reserved Instances	Purchase 1-year RI for RDS and consistent workloads	30-40% on databases	Medium
Auto-scaling	Scale down to minimum during off-hours (10 PM - 6 AM)	40% on compute	High
Data Lifecycle	Move old telemetry to cold storage after 30 days	50% on storage	Medium
Savings Plan	AWS Compute Savings Plan for predictable usage	25% across compute	Medium
Remove Shield Advanced	Not needed for university pilot phase	$3,000/month	High

Optimized Monthly Total: ~1,900 –2,200/month (after removing Shield Advanced and applying spot/reserved instances)

Optimized Annual Total: ~22,800 –26,400/year

Deliverable 8: Traceability Matrix

Pain Point to Technology Solution Traceability

Pain Point ID	Pain Point Description	Technology Solution	Component	Justification
P1	Procurement-to-inventory delay (3-5 days)	RDS + MSK + Debezium	Data Layer	CDC enables real-time asset creation from procurement events
P3	Poor mobile experience for scanning	CloudFront + S3 + React Native	Delivery	CDN ensures fast mobile app asset delivery; React Native enables offline scanning
P5	Software license over-purchasing	RDS + Elasticsearch	Search/Analytics	Fast search enables license usage tracking and compliance reporting
P7	Asset non-return after offboarding (30% loss)	Temporal Cloud + RDS	Workflow	Durable workflow ensures offboarding steps complete with retries and escalation

Pain Point ID	Pain Point Description	Technology Solution	Component	Justification
P9	Finance maintains separate spreadsheets	RDS + MWAA	Batch	Automated monthly exports to ERP via Airflow DAGs
P12	Audit findings on inventory accuracy	RDS + Neo4j + Elasticsearch	All	Complete audit trail and relationship mapping for auditors
P15	Predictive maintenance failures	InfluxDB + SageMaker	AI/Telemetry	Time-series database enables ML-based failure prediction from telemetry data

TO-BE Process to Technology Support Traceability

TO-BE Process	Technology Support	Components	Deployment
1.0 User Sync	HRIS webhook → API → RDS	API on EKS, RDS	Private subnet
2.0 Procurement Intake	Procurement API → MSK → RDS	MSK, Debezium, RDS	Private + Data subnets
3.0 Telemetry Ingestion	MDM (JAMF/Intune) → Telemetry Service → InfluxDB	EKS, InfluxDB Cloud	Private subnet + SaaS

TO-BE Process	Technology Support	Components	Deployment
4.0 AI Analytics	InfluxDB → SageMaker → Kafka	SageMaker, MSK, EKS	AI/ML subnet
5.0 Network Mapping	Network Scanner → Neo4j	Neo4j Aura, EKS	Private subnet + SaaS
6.0 License Management	RDS → Elasticsearch	Elastic Cloud, RDS	Private subnet + SaaS
7.0 Asset Retirement	API → Temporal → RDS	Temporal Cloud, EKS, RDS	Private subnet + SaaS

Data Architecture to Technology Mapping

Data Architecture Element	Technology Solution	Component	Justification
Enterprise Asset Entity	PostgreSQL (RDS) + Asset domain model	Core Database	ACID compliance for asset transactions
Snipe-IT Core Tables	RDS PostgreSQL	Core Database	Native Snipe-IT support
Custom Assignment History	RDS + Assignment Repository	Data Layer	Complete audit trail for compliance
Telemetry Time-Series	InfluxDB Cloud	Time-Series Database	Optimized for high-volume telemetry writes

Data Architecture Element	Technology Solution	Component	Justification
Graph Network Relationships	Neo4j Aura	Graph Database	Efficient relationship traversal
CDC Event Stream	Debezium + MSK	Integration Layer	Real-time change propagation
Metadata Document Store	MongoDB Atlas	Document Store	Flexible schema for warranty docs
Full-Text Search	Elastic Cloud	Search Engine	Fast asset lookup and discovery

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Deliverable 9: Executive Summary Presentation (Slides Outline)

Slide 1: Title Slide

- PUP IT Asset Management System
- Technology Architecture Design
- Team Name, Date

Slide 2: Executive Summary

- Problem: Manual processes, asset loss, audit failures
- Solution: Cloud-native polyglot architecture on AWS
- Key Outcomes: 99.95% availability, 90% operational hours saved

Slide 3: Deployment Architecture

- AWS Singapore region (ap-southeast-1)
- EKS for containers, RDS for PostgreSQL, SaaS for polyglot databases

- Multi-AZ for all critical components

Slide 4: Network Security

- VPC with 5 subnets (public, app, data, SaaS, batch)
- Defense in depth with security groups
- Encryption at rest and in transit

Slide 5: Technology Stack Highlights

- Compute: AWS EKS with Kubernetes 1.28
- Core DB: RDS PostgreSQL
- Polyglot: MongoDB Atlas, InfluxDB Cloud, Neo4j Aura
- AI/ML: SageMaker for demand forecasting & predictive maintenance
- Integration: MSK Kafka, Debezium, Temporal Cloud

Slide 6: Managed Services Strategy

- 10 managed services selected over self-hosted
- Cost premium: ~$175/month
- Operational hours saved: 90 hours/month (~$9,000 value)

Slide 7: Scalability & HA

- Horizontal auto-scaling for API (3-20 pods)
- Multi-AZ for all data services
- RTO: < 1 hour for critical systems
- SLA: 99.95% for core services

Slide 8: Cost Summary

- Base monthly cost: $4,927
- Optimized monthly cost: $1,900 - 2,200$
- Annual optimized: $22,800 - 26,400$

Slide 9: Traceability to Pain Points

- P1 (Procurement delay) → CDC with Debezium + Kafka

- P7 (Asset non-return) → Temporal workflows
- P15 (Predictive failures) → InfluxDB + SageMaker

Slide 10: Next Steps

- Phase 1: Deploy core infrastructure (Terraform)
- Phase 2: Implement CDC pipeline
- Phase 3: Train and deploy AI models
- Phase 4: Full cutover and optimization

Conclusion

This sample solution demonstrates a complete technology architecture for a university IT asset management transformation. The architecture leverages AWS cloud services, polyglot persistence, event-driven AI, and infrastructure as code to address specific business pain points while maintaining high availability, security, and cost efficiency.

Key Decisions:

- AWS over Azure/GCP for SageMaker and mature services
- Managed services over self-hosted for operational efficiency
- Polyglot persistence to match database to data requirements
- Event-driven AI for scalable predictive maintenance
- Terraform for infrastructure as code

Estimated Monthly Cost (Optimized): $1,900 - 2,200$
Estimated Annual Cost (Optimized): $22,800 - 26,400$
Operational Hours Saved: 90 hours/month
RTO: < 1 hour for critical systems
RPO: < 5 minutes for critical data

End of Activity 5 Sample Solution